

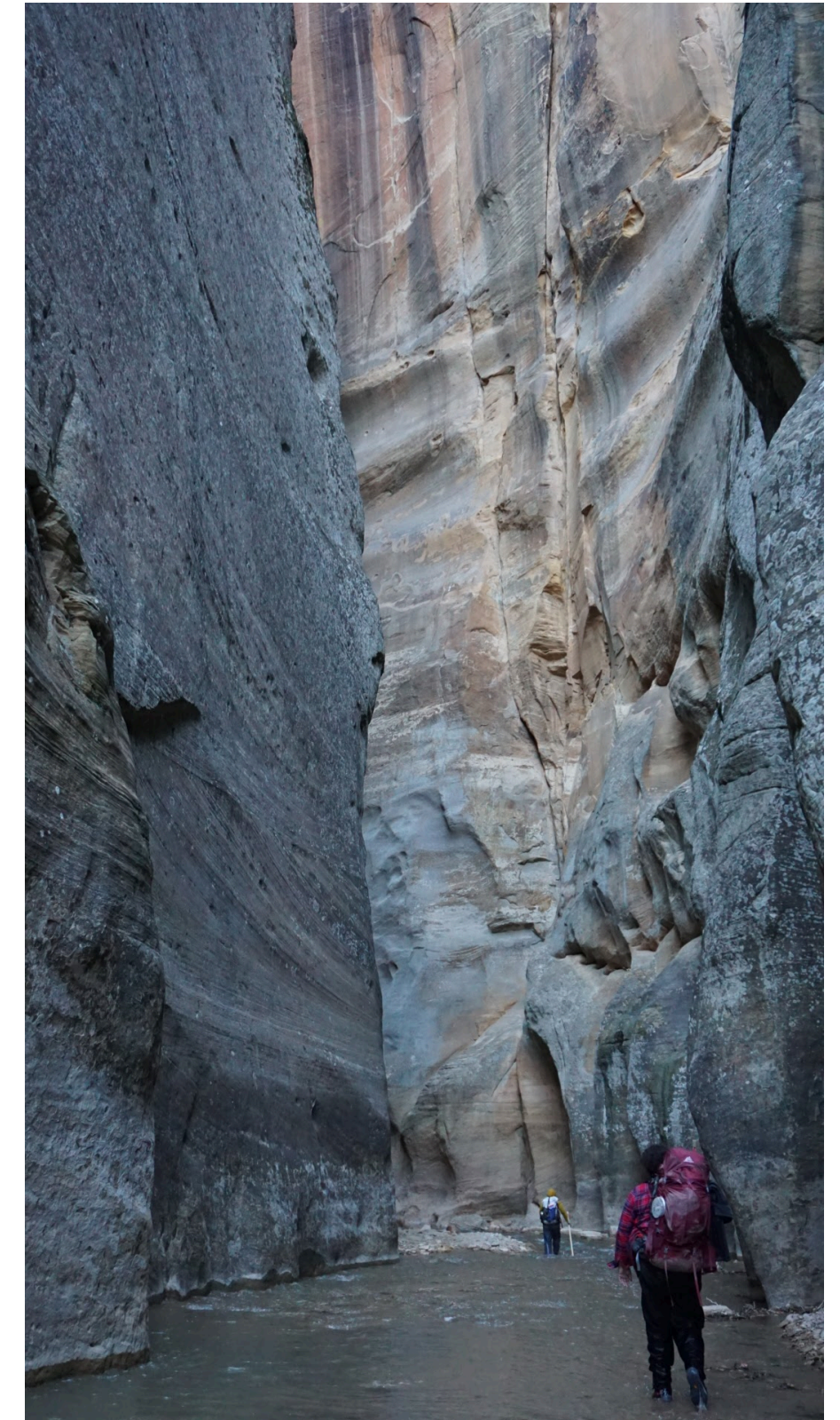
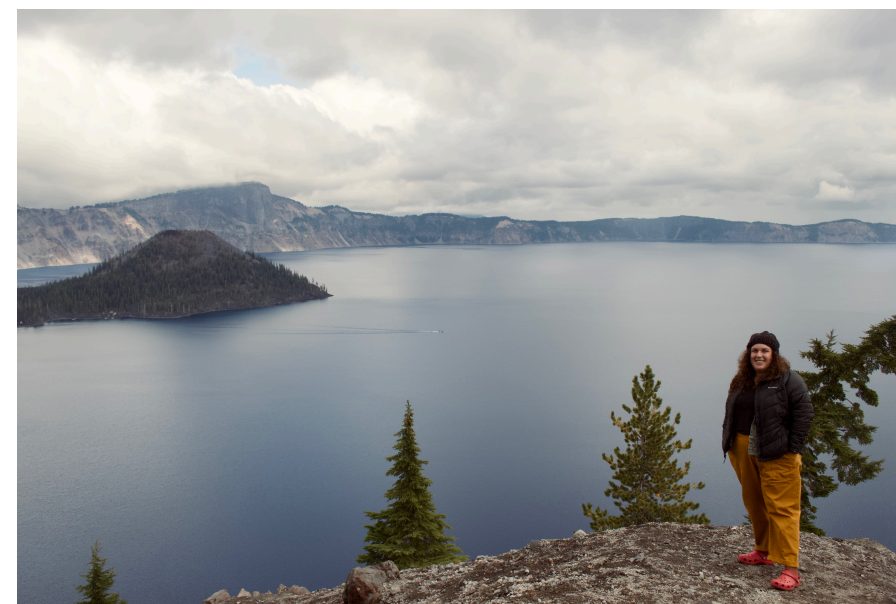
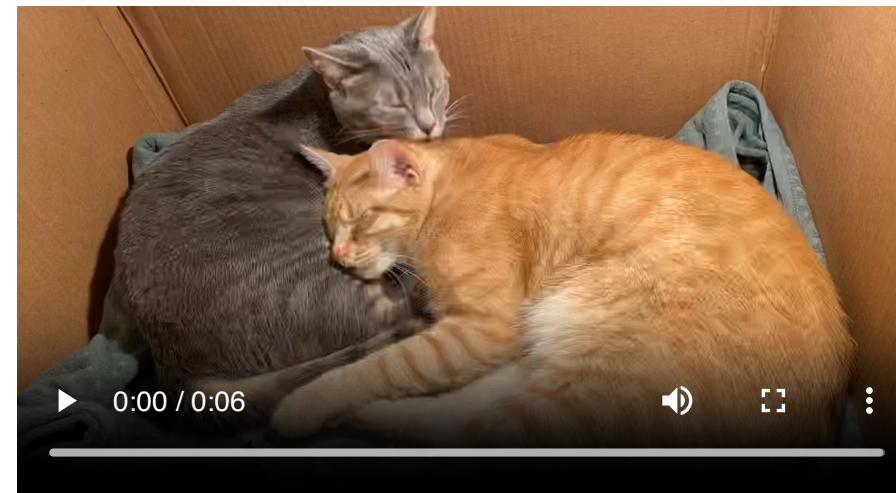
Welcome to BSTA 550!

Nicky Wakim

2026-09-28

Nicky Wakim (she/her)

- Call me “Nicky,” “Dr. W,” “Professor Wakim,” or any combo!
- Assistant Professor of Biostatistics
- Grew up in DC area (Virginia side!)
- Moved here from Michigan around 2 years ago
- Two sweet kitties
- Volleyball, pickleball, ceramics, strolling around my neighborhood
- But also sleeping, TV, and reading
- Proud plant mamma
- *A few other things about myself that I will share non-publicly*



Pride yourself in learning things, not knowing things

Some important tasks

- Star the class website: https://nwakim.github.io/BSTA_550_26F/
- Complete Homework 0 by this Thursday at 11pm!
 - Includes office hours set up, attendance preference, and homework due date decision
 - Think about what day of the week you would like your homeworks due
- Highly suggest that you make an appointment with a learning specialist through [Student Academic Success Center](#)!

Let's visit the website: Homepage

 Introduction to Probability

SCHEDULESYLLABUSINSTRUCTORSHOMEWORKQUIZZESPROJECT



BSTA 550: Introduction to Probability

Fall 2026

Welcome to Biostatistics and Probability!

This course is an introduction to the methods and concepts of probability theory, including conditional probability and independence, combinatorics, discrete and continuous random variables, probability distributions, joint distributions, expectation, transformations of random variables, and the central limit theorem. Students will apply probability concepts and develop statistical reasoning through simulations in R.

[OneDrive Folder](#)[Syllabus](#)[Schedule](#)

Instructor

 [Dr. Nicky Wakim](#)

 Vanport 622A

 wakim@ohsu.edu

Office Hours

[Nicky](#)

 TBD

TA

 TBD

Course details

 Mondays, Wednesdays

 09/28 - 12/07

 10:30 AM - 12 PM

 In-person, VPT 264

Contacting me

 E-mail is the best way to reach me. I try to respond to all course-related e-mails within 24 hours, Monday-Friday.

 [View the source on GitHub](#)

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Welcome

Let's visit the website: Syllabus (1/2)

- Course learning objectives
- Textbook online! (different than last year)
- Resources: PennState STAT 414 site!
- R: you will get a lot of help in BSTA 511 and we will use some!
- Assessments and grade breakdowns
 - Mostly homework + quizzes
- Feedback from you to me: in the form of exit tickets, midterm feedback, and final course eval
- How to succeed in this course: resources and assignments explained
- Late work policy / Attendance policy
- ChatGPT and other AI technology
- Course expectations: a few ways that I will show you respect and commitment to you as students
 - And a few ways I expect from you!
- Communicating with me: give me 24 hours to reply M-F
 - I try really hard to keep emails from taking over my life

Let’s visit the website: Syllabus (2/2)

 Introduction to Probability

SCHEDULESYLLABUSINSTRUCTORSHOMEWORKQUIZZESPROJECT

BSTA 550 Syllabus

MODIFIED
September 25, 2026

Key Course Info

- **Late work or a closed Brightspace assignment?** Email your work to **both me and the TA** (addresses on the [instructor page](#)).
- **Need an extension?** Email me.
- **Homework:** turn it in any time before the course ends. On-time homework gets feedback; late homework does not.
- **Project checkpoints:** you get ONE no-questions-asked, 3-day extension.
- **Attendance:** try to come to class in-person. Recordings will be available.
- **Used AI on an assignment?** Add an AI Use Statement at the top (see the [Generative AI Policy](#)).
- Class ends December 07, 2026. **All coursework MUST be completed by December 11, 2026.**
- Questions or concerns anytime? Use the [Ongoing Course Feedback](#) form.

Description

Welcome to BSTA 550! This course introduces the methods and concepts of probability theory: conditional probability and independence, combinatorics, discrete and continuous random variables, probability distributions, joint distributions, expectation, transformations of random variables, and the central limit theorem. You’ll apply these concepts and build statistical reasoning through simulations in R.

Course Learning Objectives

By the end of this course, you should be able to...

1. Assign probability to a chance event using concepts of probability (including fundamental axioms, properties, and counting).
2. Compute probabilities for *discrete random variables* (including random variables following Bernoulli, binomial, geometric, and Poisson distributions).
3. Compute probabilities for *continuous random variables* (including random variables following Normal, Gamma, and Beta distributions).
4. Perform statistical computations and simulations using R.

Let's visit the website: Schedule (1/2)

Let's visit the website: Schedule (2/2)

	Key Info	I will post announcements and other important class related info here. For example, if I change a due date or discuss a common mistake in homework, I will put it here.
	Slides HTML	These are the basic slides that will open in your browser.
	Slides PDF	These are the slides in pdf form for easy note taking. I'm not always the best at posting these before class, so make sure you know how to save your own copy of pdf slides!
	Slides Notes	These are the annotated slides in pdf form. In class, I add my own notes to slides. After class, I will post them here.
	Exit tix	These are links to that day's exit ticket.
	Recording	I record our classes. This will be a link to the OneDrive folder containing this recording.
	Muddy Points	You will have a chance to ask questions about class in your exit tickets. If I notice a trend in confusion, I will add explanations to these "Muddy Points"

Let’s visit the website: Search

 Introduction to Probability

SCHEDULESYLLABUSINSTRUCTORSHOMEWORKQUIZZESPROJECT

Schedule

Week	Date	Key Info	Exit tix	Recording	Lesson	Topic	TB	Slides HTML	Slides PDF	Slides Notes	Muddy Points
1	09/28				0	Welcome					
					1	Introduction to Probability	1.1, 2.1, 2.2, 2.4.1				
	09/30				2	Introduction to Simulations	2.3, 2.5				
10/01						HW 0 due 11pm					
2	10/05				3	Language of Probability	2.2, 2.4				
	10/07				4	Rules of Probability	2.7, 3.3-3.5				
	10/11						HW 1 due 11pm				
	10/11						Checkpoint 0 due 11pm				
3	10/12				6	Async optional video: Calculus Review					
	10/14				5	Virtual class: Equally Likely Outcomes	3.6				
	10/18						Checkpoint 1 due 11pm				
4	10/19				7	pmfs	4.1, 4.2				
					8	pdfs	4.3				

Let’s visit the website: Homework!

 Introduction to Probability

SCHEDULESYLLABUSINSTRUCTORSHOMEWORKQUIZZESPROJECT

Homework

MODIFIED
September 26, 2026

Homework	Assignment	Assignment due (@11pm)	Answers	Solutions & videos
0		10/01		
1		10/11		
2		10/25		
3		11/01		
4		11/08		
5		11/22		
6		12/06		
7 (optional)		12/10		

File Naming

For HW **Assignments**, please use the following file naming: **HW01_LastName_FirstInitial**

- For homeworks without R, this should be a **pdf** file
- For homeworks with R, this may be a **pdf** file (with the code in the pdf) or an **html** file

Rubric

Decision on Homework due dates

- I have some set due dates in the schedule
- Please look at your other classes, your calendar, etc
- Consider what day of the week you would like to turn in your assignments
- Question in HW 0 to cast your vote and share your opinion

Structure for this course

- Learning the basic tools to understand statistics
 - This is the first quarter that I am adding simulations
 - So I rearranged a lot of the topics!
- It is going to feel useless at times, but I swear it is not!
- This class will help you build a toolbox that allows you to analyze data while understanding the inner theory at play
- You can use probability and simulations to change your analysis as you need

What we will cover

